Princy Ranaivomanana, PhD

Data Science & Research Expertise | ML Engineer | MLOps

Netherlands · Open to EU/International Roles

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PROFILE

My name is Princy. I am originally from Madagascar. I have recently completed a PhD in Astronomy. I chose this topic as I love turning (observational) data into insights and improving our knowledge of the universe. I have experiences in analysing and processing large datasets (structured and unstructured), modelling data to uncover trends and patterns, and developing (un)supervised machine learning models, which I acquired during my PhD research. Besides, I am flexible, an independent and team player, always curious, and eager to learn new skills.

TECHNICAL SKILLS

- Python for data science
- SQL: Querying data from relational databases
- Machine learning: Feature engineering, classification/predictive modelling, clustering
- Deep Learning: Building generative AI models (GANs)

WORKPLACE SKILLS

- Communication (reporting and presentations)
- Teamwork and collaboration
- Problem-solving

DATA ANALYTIC TOOLS

- Pandas, Scikit-learn, Matplotlib, Jupyter Notebook
- IBM Watson Knowledge Studio
- GitHub, Git

PROFESSIONAL EXPERIENCE

1. **Industry Collaboration Project – Nutrico** *16-18 Feb 2026*

Annual modelling camp organised by the Mathematics Department, Radboud University.

Collaborated within a multidisciplinary team of researchers to develop machine learning solutions for genotype determination at the Nutrico (Trouw Nutrition) company, reducing processing time from one week to several minutes.

2. **PhD Researcher (Astronomy)** *Radboud University & KU Leuven | Oct 2021 – Sep 2025*

a) Project leader for a machine learning effort involving the classification of massive time-series data.

- Outcome: Distinguished various clusters of objects using t-SNE/UMAP and Gaussian mixture models.
- Deliverable: Two peer-reviewed publications in a high-index journal.

b) Contributed to the solution of periodicity search in irregularly sparsely sampled time series.

c) Authored four peer-reviewed publications.

3. **Teaching Assistant** *Radboud University & KU Leuven | Oct 2021 – Sep 2025*

- Tutored courses on data analysis and astronomical instrumentation
- Mentored two bachelor research projects (two students per project)

EDUCATION

PhD, Astronomy & Astrophysics (Dual Degree): Radboud University & KU Leuven | 2021 – 2025

MPhil, Astronomy: University of Manchester, United Kingdom | 2020 – 2021

MSc, Astronomy: University of Antananarivo, Madagascar | 2016 – 2017

BSc, Physics: University of Antananarivo, Madagascar | 2012 – 2015

PRIZES AND GRANTS

- Receipt of the LKBF grants for attending conferences (2023/2024/2025)
- Recipient of the UK's Newton Fund/DARA Big Data Masters Bursary (2020)
- DARA travel grant − African Radio Interferometry Winter School − South Africa (2019)
- Receipt of SKA/SARAO grant to attend the 2nd Big Data Africa School − South Africa (2018)
- IAU travel grant − The 39th International School for Young Astronomers (ISYA) − Ethiopia (2017)

RELEVANT CONFERENCES ATTENDED & CONTRIBUTIONS

Invited speaker:

- Invited talk at the Hamburg Observatory Colloquium (November 2025)
- Talk title: **Unsupervised machine learning** for variability analysis in time-domain astronomy

Conference talk:

- Conference: The 12th international meeting on hot subdwarfs & related objects (May 2025)
- Talk title: Unravelling Hot Subdwarfs and Binaries Variability with Gaia DR3 and **Machine Learning**

Machine learning conference: Oxford machine learning school 2023 & 2025

RELEVANT CERTIFICATION

- Data Fundamentals (issued by IBM SkillsBuild) Online Certificate
- Machine Learning in Production (issued by DeepLearning.AI) Online Certificate
- Advanced Machine Learning and Signal Processing (issued by Coursera) Online Certificate
- Data Analysis with Python (issued by Coursera) Online Certificate
- Exploratory Data Analysis for Machine Learning (issued by Coursera) Online Certificate

SELECTED STRENGTHS FOR INDUSTRY

- Translating complex problems into understandable solutions
- Quick and flexible learner
- Working with imperfect, real-world data at scale